

List of minimum Tools & Equipments for a batch of 30 students			
Programme Title : KGCE in Mechanical Engineering		Notional Hours: 2880	
Sl No	Name of the Tool & Equipment	Specification	Quantity
GENERAL SHOP TOOLS			
General Tools			
1	Stainless Steel Rule (with Metric & British graduation)	300 mm	15 Nos.
2	Try Square	150 mm blade	15 Nos.
3	Scriber	150 mm	15 Nos.
4	Hammer Ball peen	450 g with handle (1lb)	6 Nos.
5	Bench vice	6" heavy duty	15 Nos.
6	Number punch	6 mm	2 set
7	Letter punch	6 mm	2 set
8	Vernier caliper	150 mm	4 nos.
9	Flat file	6"	15 Nos.
10	Centre Punch	10 mm - Length 120 mm	6 Nos.
11	Divider spring type	150 mm	6 Nos.
12	File flat - bastard	300 mm	6 nos.
13	SS measuring tape	5 meter flexible in case	2 nos.
14	Spanner set (Double end)	6mm to 34mm	2 set
15	Chipping hammer		2 Nos.
16	Hammer Ball peen	1kg with handle	2 Nos.
17	Angle grinder		1 no.
18	Bench grinder fitted with fine grain size silicon carbide green grinding wheel	150 mm dia	1 no.
19	Anvil	12 sq inches working area with stand	1 no.
20	V-Block	150x100x100 mm	1 no.
FITTING			
1	Caliper inside spring type	150 mm	6 Nos.
2	Caliper Outside spring type	150 mm	6 Nos.
3	Dot Punch	100 mm	6 Nos.
4	File flat - Second cut	250 mm	15 Nos.
5	File flat - Smooth	250 mm	15 Nos.
6	File flat second cut safe edge	300 mm	6 nos.
7	File half round - Second cut	300 mm	12 Nos.
8	File half round bastard.	250 mm	6 nos.
9	File round - Second cut	300 mm	12 Nos.
10	File round bastard	250 mm	6 nos.
11	File triangular - Smooth	300 mm	12 Nos.
12	File triangular - second cut	200 mm	6 nos.
13	File knife edge smooth	150 mm	3 nos.
14	File feather edge smooth	150 mm	3 nos.
15	File square second cut	250 mm	6 nos.

16	File Swiss type needle	Set of 12, Length = 150 mm	1 set
17	File hand second cut	150 mm	6 nos.
18	Hammer Ball peen	220 g with handle (1/2 lb)	6 Nos.
19	Chisel cold	Flat 20mm x 150mm HCS	12 Nos.
20	Hacksaw frame fixed type	300 mm	15 Nos.
21	Safety goggles		15 Nos.
22	Stud Extractor EZY - out	Set of 8	2 sets
23	Combination Set	300 mm	2 nos.
24	Micrometer outside.	0 - 25 mm	2 nos.
25	Micrometer outside.	25 - 50 mm	2 nos.
26	Micrometer outside.	50 - 75 mm	2 nos.
27	Micrometer inside with extension rods.	Accuracy 0.01 mm with extension rods up to 150 mm	1 no.
28	Screw pitch gauge Metric	0.25 to 6 mm	1 no.
29	Wire gauge, metric standard.		1 no.
30	Spanner- Double Ended	6x7, 8x9, 10x11, 12x13, 14x15, 16x17, 18x19, 20x22	1 set each
31	Spanner adjustable	150 mm	2 nos.
32	Interchangeable ratchet socket set	12 mm driver, sized 10-32 mm set of 18 socket & attachments.	1 set
33	Double Ended tubular Box spanner set with Tommy bar	A/F 6-25 mm set of 10 Tommy Bar Dia. 6, 8, 10, 12, 14, 16	1 set
34	Clamp "C"	100 mm	2 nos.
35	Clamp "C"	200 mm	2 nos.
36	Hand Reamer set (Taper pin straight flute)	Nominal Dia. 6, 8, 10, 12, 16mm	1 set
37	Machine Reamer parallel (Helical flute)	12 - 16mm set of 5.	1 no.
38	Scraper flat	150 mm	6 nos.
39	Scraper triangular	150 mm	6 nos.
40	Scraper half round	150 mm	6 nos.
41	Chisel cold crosscut & diamond point.	9 mm X 150 mm	6 each
42	Chisel cold flat	9 mm X 100 mm	6 nos.
43	Chisel cold round nose	9 mm X 100 mm	6 nos.
44	Drill chuck with key	12 mm.	1 no.
45	Pipe wrench	400 mm	1 no.
46	Pipe vice	100 mm	1 no.
47	Dowel pin (straight)	Dia. -1" Length -4" (Mat: Stainless Steel)	1 each
48	Standard Tap screws	M3, M4, M5, M6, M8, M10, M12, M14, M16	1 each
49	Lapping plate	Dia. -6"	2 each
50	Medium carbon Heat treated alloy steel Metric Studs and bolts along with nuts (for display) of standard length (May be manufactured in-house)	M6, M8, M10, M12, M14, M16 (Standard)	2 each
51	Caps screws	M6, M8, M10, M12	2 each

52	Drill gauges	Letter drill gauge (A to Z), Number drill gauge (1 to 60), Metric drill gauge (1.5mm to 12.5mm, 30 holes)	2 nos.
53	Cast Iron Globe Valve (Flanged type)	150NB, Class# 150 Flange:ANSI125-B16.1	2 nos.
54	C.I. Sluice / Gate valve (flanged type)	150NB, Class# 150 Flange:	2 nos.
55	Bolt & Nut with washer (May be manufactured in-house)	M20x2.5x90Long (part thread - ex.Head)	15 nos.
56	Pipe threading die with handle	Ratchet type Die head of 1/2", 3/4" and 1"	2 nos.
57	Jigs & Fixture (sample)-For demonstration (May be manufactured in-house)		1 no.
58	Pulleys (for V-belt or Flat belt)	to fit on 50mm dia. Shaft with key slot	1 no.
59	Steel keys (May be manufactured in-house)	to fit with key slot of shaft & pulley	2 nos.
60	Damaged old spur gear	to fit 50mm dia. Shaft	2 nos.
61	V-belt and Flat belt	to fit on pulley	1 each
62	Packing gasket	PTFE gasket roll small size	1 no.
63	Washer, clutch, keys, jib, cotter & circlip	minimum 25mm size, carbon steel material	2 each
64	Hollow punch	Straight Shank Hollow Punch Sets 5-12mm	1 set
65	Drill Drift (May be manufactured in-house)	200mm hardened and black finish	2 nos.
66	Bearing different types	Each type of diameter 25mm (min.)	1 each
67	Lifting sling	8mm Nominal Dia. Single leg sling	2 nos.
68	Bearing extractor	Universal gear puller 2 or 3 jaws adjustable	1 no.
69	Pulley extractor		1 no.
70	Slip Gauge as Johnson metric set	87 Pieces Set	1 Set
71	Gauge snap Go and Not Go	25 to 50 mm by 5 mm, Set of 6 pieces	1 Set
72	Gauge plug	Single ended 5 to 55 by 5 mm. Set of 11 pcs.	1 Set
73	Gauge telescopic set	8 - 150 mm	1 no.
74	Dial test indicator on stand	0.01 mm least count	1 no.
75	Sine bar	125 mm	1 no.
76	Dial Vernier caliper. (Universal type)	0 - 300 mm	1 no.
77	Screw thread micrometer with interchangeable. Pitch anvils for checking metric threads 60.	0 - 25 mm, LC 0.01 mm	1 no.
78	Depth micrometer. 0-25 mm	Accuracy 0.01 mm with standard set of extension rods up to 200 mm	1 no.
79	Digital Vernier caliper.	0 - 150 mm with least count 0.02mm	1 no.
80	Digital Micrometer outside.	0 - 25 mm L.C. 0.001 mm.	1 no.
81	Comparators Gauge - Dial Indication with Stand and Bracket.	LC 0.01mm	1 no.

82	Engineer's try square (knife-edge)	150 mm Blade	1 no.
83	Surface roughness comparison plates	N1 - N12 Grade	1 Set
84	Digital Vernier caliper	0 - 200 mm L.C. 0.01 mm (Optional)	1 no.
85	Vernier Height Gauge	300 mm	1 no.
86	Surface Plate (C.I/Granite with Stand and Cover)	600 x 600 mm	1 no.
87	Marking table (Mild steel)	900x900x900 mm	1 no.
88			
89	Angle plate	150 X 150 X 250 mm	2 nos.
90	Tap & Die set	Whitworth	1 set
91	Feeler gauge	Gauge Feeler / Thickness - 0.05 mm to 0.3 mm by 0.05 and 0.4 mm to 1 mm by 0.1 mm - 13 leaves	1 set
92	Metal rack	200cm x 150cm x 45 cm	1 no.
93	First aid box		1 no.
94	Steel lockers	with 8 Pigeon holes	2 nos.
95	File card / Wire Brush	3"x5" size, brass or steel wire	6 nos.
96	Oil Can	250 ml	2 nos.
97	Pliers combination insulated	150 mm	2 nos.
98	Drill twist straight shank	3 mm to 12 mm by 0.5 mm H.S.S	1 no.
99	Drill twist Taper shank	8 mm to 20 mm by 0.5 mm H.S.S	1 no.
100	Portable hand drill (Electric)	0 to 13 mm Capacity	1 no.
101	Screw driver	150mm insulated flat type	6 Nos.
SHEET METAL			
1	Straight Snips	8"	15 Nos.
2	Bend Snips	8"	6 Nos.
3	Groover	2"	6 Nos.
4	Standard Wire Gauge		2 Nos.
5	Mallet	2"	15 Nos.
6	Hatchet stake	6"	2 nos.
7	Funnel stake	6"	2 nos.
8	Conductor stake	6"	2 nos.
9	Copper smith stake	6"	2 nos.
10	Rivet set		2 nos
11	Wooden handle forged Soldering Iron copper bit.	230V, 250 W, 350 gm	2 nos.
12	Blow Lamp	0.5 litre	2 nos.
13	Trammel	300 mm	1 no.
14	Pocker		2 nos.
15	Prick punch	100 mm	4 nos.
16	Flat headed hammers with handle		6 nos.
17	Planishing hammer		4 nos.
18	Snip bent Left Cut	250 mm	4 nos.
19	Metal rack	200cm x 150cm x 45 cm	1
20	First aid box		1
21	Steel lockers	with 8 Pigeon holes	2
WELDING			

1	Hack saw frame	Fixed (300 mm)	15 Nos.
2	File flat	350 mm bastard	15 Nos.
3	Chisel cold	flat 19mm x 150mm	6 Nos.
4	Welding helmet fiber		6 Nos.
5	Welding hand shield fiber		6 Nos.
6	Wire brush		6 Nos.
7	Tongs (Flat)	350 mm	6 Nos.
8	Tongs (Round)	350 mm	6 Nos.
9	Screw driver	300mm blade and 250mm blade	1 each
10	Magnifying glass	100 mm dia	2 nos
11	Universal weld measuring gauge		2 nos
12	Earth clamp	600A	6 nos.
13	C- Clamp	10 cm and 15 cm	2 each
14	Sledge hammer	Double faced, 4 kg	2 nos.
15	Electrode holder	600 A	6 nos.
16	Metal rack	200cm x 150cm x 45 cm	1
17	First aid box		1
18	Steel lockers	with 8 Pigeon holes	2
19	File card		2 nos.
20	Gas lighter		2 Nos.
21	Trolley for gas cylinders.		1 No
22	Spindle key for cylinder valve.		2 Nos.
23	Welding torches.	Nozzles no. 1, 2, & 3	1 Set.
24	Welding goggles		4 Pairs.
25	Welding helmet with coloured flame retardant glass		2 Nos.
26	Tip cleaner		5 Sets.
27	Auto darkening welding helmet		2 nos.
28	Swage block		1 no
29	Die penetrant testing kit		1 set
30	Magnetic particle testing kit		1 set
TURNING			
1	Micrometer	0-25mm	2 Nos.
2	Chuck Key		2 Nos.
3	Double End spanner set	6mm to 34mm	2 Nos.
4	Surface Gauge		2 Nos.
5	Centres		2 Nos.
6	Inside Caliper		2 Nos.
7	Outside Caliper		2 Nos.
8	Screw Pitch Gauge		2 Nos.
9	Different Profile gauges (Plate type) - For demonstration	Metric standard	2 nos.
10	Knurling tool (Diamond, straight & Diagonal)		1 each
11	Lathe tools H.S.S. tipped set.		2 nos.
12	Lathe tools bit.	6 mm x 75 mm HSS/Carbide	2 nos.
13	Lathe tools bit.	8 mm x 75 mm HSS/Carbide	2 nos.

14	Lathe tools bit.	10 mm x 75 mm HSS/Carbide	2 nos.
15	Surface Plate	1000 x 1000 mm	1 no.
16	Universal Surface gauge	250 mm arm	2 no.
17	Hammer ball peen	750 gm with handle	2nos.
18	Digimatic Electronic Vernier Caliper	inch and mm 8"/200 mm. LCM 0.005"/0.001 mm	1 no.
19	Digimatic electronic outside Micrometer	(0 to 25 mm & 25 to 50 mm) LC 0.001 mm.	1 no.
20	Morse Taper	Sleeves No. 0-1, 1-2, 2-3, 3-4, 4-5	1 set
21	Lathe Mandrels (Diff. Types)		1 set
GENERAL EQUIPMENTS & FURNITURE			
FITTING			
1	Metal cut-off machine	14" dia wheel	1 no.
2	Power hack saw machine	14"	1 no.
3	Portable drilling machine (with drill bit set)	Capacity 6mm	1 no
4	Drill machine pillar type-motorized	up to 12 mm. Cap, Motor Capacity-0.75 KW	1 no.
5	Bench working	2400 x 1200 x 900 mm	6 nos.
6	Almirah	1800 x 900 x 450 mm	2 nos.
7	Lockers with 8 drawers (standard size).	One locker for each trainee	4 nos.
8	Metal rack	1820 x 1820 x 450 cm	1 no.
9	Instructor Table		
10	Instructor Chair		
11	Black board with Stand		1 no.
12	Fire extinguisher (Common to all sections)	Foam type and CO2 type	4 nos (Min)
13	Fire buckets with stand (Common to all sections)		4 Nos.
SHEET METAL			
1	Shearing Machine		1 no.
2	Sheet Bending/folding machine		1 no.
3	Drill machine pillar type-motorized	up to 12 mm. Cap, Motor Capacity - 0.75 KW	1 no.
WELDING			
1	Welding transformer	with all accessories (400A, OCV 60-100V. 60% duty cycle)	2 set
2	Welding transformer or Inverter based welding machine	with all accessories (300A, OCV 60-100V. 60% duty cycle)	1 set

3	DC arc welding rectifier set with all accessories	400A, OCV 60-100V. 60% duty cycle	1 set
4	AC/DC GTAW welding machine	with water cooled torch 300A, Argon regulator, Gas hose, water circulating system and standard accessories	1 set
5	Oxy - acetylene gas welding set equipment with hoses, Oxygen & Acetylene cylinders, regulator and other accessories.		2 Set
6	Gas welding table with positioner with Fire Bricks	900 X 600 X 750 mm	1 No
7	Welding torch tips of different sizes for Oxy - acetylene gas welding	To fit nozzle no. 1, 2, & 3	1 Set
8	Bench Grinder		2 nos.
9	Metal cut-off machine	14" dia wheel	1 no.
10	Suitable arc welding table with booth	with positioner	6 nos.
11	Hand shearing machine	cut 6mm sheets and flats	1 no.
12	Portable drilling machine (with drill bit set)	Capacity 6mm	1 no
13	Oven for electrode drying	0 to 350 degree C, 10kg capacity	1 no
14	Welding simulator for SMAW/GTAW/GMAW		1 no
TURNING			
1	Lathe S.S. & S.C. (All geared head stock) with all accessories and attachments with minimum specification as:	150 mm center height, to admit 750 mm between centers. Machine to be motorized and supplied with coolant installation, 4-jaw Independent chuck 150 mm, 3-jaw self-centering chuck 150 mm, fixed steady, traveling steady, face plate, driving plate, 4-way tool post, quick change gear box for Metric or British threads, live and dead centers with taper attachments, Motor Capacity - 5.5 KW	2 nos.
2	Grinding machine pedestal type	D.E. 200 mm dia. Wheel with wheel guard and vision, Motor Capacity -0.75 KW	1 no.
3	Drill machine pillar type-motorized	up to 12 mm. Cap, Motor Capacity -0.75 KW	1 no.
4	Power saw machine – hydraulic feed system	400 mm. Blade size, Motor Capacity -0.75 KW	1 no.
CONSUMABLES			
1	Hand gloves	14" - Leather	6 pair.
2	Hand gloves	8" Cotton	6 pair.
3	Apron	Leather	6 Nos.

4	SS Wire brush		6 Nos.
5	Leather hand sleeves	16"	6 Nos.
6	Safety boots for welders		6 pairs
7	Leg guards	Leather	6 pairs
8	Rubber hose clips	1/2"	6 nos
9	Arc welding cables multi cored copper	400/600amps as per BIS	45 mts each
10	Arc welding single coloured glasses	108mm x 82mm x 3mm DIN 11A & 12A	30 nos
11	Arc welding plain glass	100mm x 82mm 3 mm	30 nos
12	AG4 grinding wheel		As required
13	Cutting tool		As required
14	Electrode		As required
15	MS Flat		As required
16	MS Rod		As required
17	GI Sheet		As required
18	Lubrication Oil		As required
19	Cotton Waste		As required

Common facility for all programmes in an Institution			
CLASS ROOM FURNITURE			
1	Instructors table		1 no.
2	Instructors chair		1 no.
3	Students chair with writing pads/bench-desk arrangement		for 30 students
4	Almirah		1 no
5	Black/White board	1200 x 900 mm	1 no.
6	Instructors Laptop with internet facility	Windows with MS office package	1 no.
7	LCD Projector		1 no
IT LAB *			
1	Table & Chair for Instructors		Sufficient
3	Students chair with writing pads/bench-desk arrangement		for 30 students
4	Steel almarah / cup board		1 no.
5	Black/White board	1200 x 900 mm	1 no.
6	Desktop with internet facility	Windows with MS office package (Latest configuration)	10 nos.
7	LCD Projector		1 no
8	Printer	A4 Size	1 no.
9	Computer table		Sufficient number for arranging 10 computers

* Common for up to 5 programmes or 5 batches in an Institution. For Institutions with more than 5 programmes or 5 batches there shall be proportionate increase in number of computers and related furniture.